

Erythrocytes

Erythrogram

- counts of red blood cells (RBC)
- RBC mass (hematocrit)
- hemoglobin concentration
- RBC indices (cell size, hemoglobin content)
- RBC size variation (red cell distribution width or RDW)
- reticulocyte counts (percentage or absolute)
- nucleated RBC count (per 100 WBC or absolute)
- RBC morphologic features, including presence of parasites.

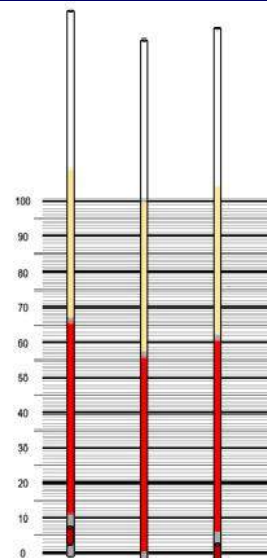
Packed cell volume (PCV)

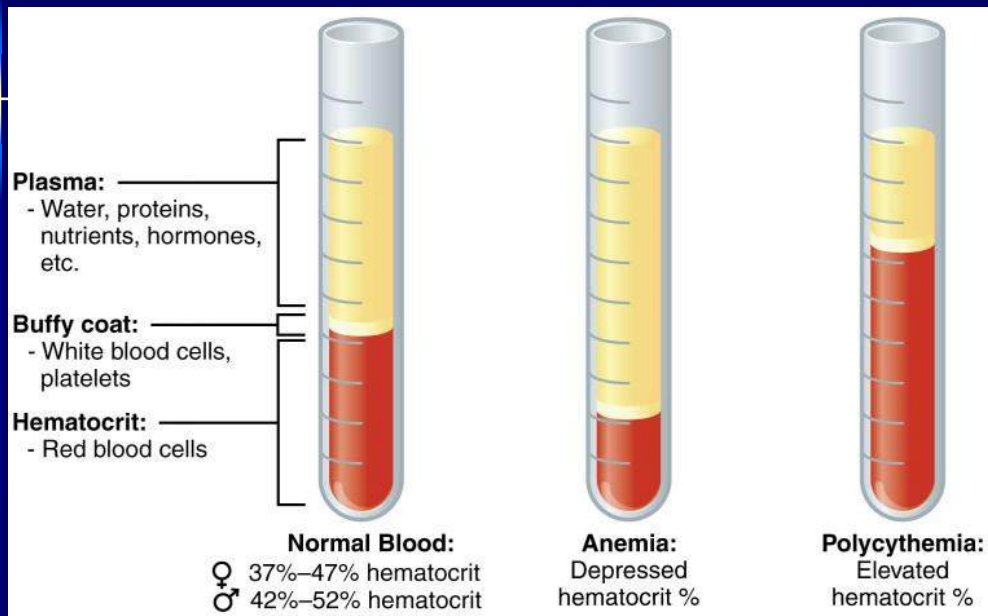
- Hematocrit is the % of blood composed of erythrocytes; to separate blood is thru centrifugation (platelets, plasma, and leukocytes are trapped between red cells)
- PCV, buffy coat and plasma
- Microhematocrit and macrohematocrit
- Red blood cell packing is species-dependent – it takes longer for ruminant RBCs to pack compared to dogs, cats and horses, Therefore, the microhematocrit tubes are spun for 10 minutes in ruminants versus 3 minutes in other species.

Hematocrit test



medicalabtechnology.com





Packed cell volume (PCV)

Plasma color and transparency

- (a) Normal plasma is clear and colorless (dog and cat) to light yellow (horse and cow)
- (b) Icteric plasma is yellow and clear
- (c) Hemoglobinemic plasma is pink to red and clear
- (d) Lipemic plasma is whitish to pink and opaque

The buffy coat zone, a white layer between the RBCs and plasma, is comprised of leukocytes and platelets. Measurement of its width has been used to estimate white blood cell counts.

Microfilaria may be detected by microscopic examination of the plasma just above the buffy coat layer.

Hemoglobin determination

- The most accurate direct indication of oxygen transport capacity of blood and approximately 1/3 of the PCV if erythrocytes are of normal size
- Acid- hematin method
- oxyhemoglobin (Tallqvist hemoglobin scale and Dare chemoglobinometer)
- Cyanmethemoglobin
- Avian erythrocytes must be lysed and the specimen must be centrifuged to remove free nuclei before Hb concentration

RBC counts

- performed with a hemocytometer, have a large degree of error; hemacytometer-derived RBC counts are of limited value, except in avian species.
- Automatic counters, if standardized for mammalian blood, allow for more accurate RBC counts
- Automated counters are not well validated for avian blood, because all nucleated cells (RBCs, WBCs, and thrombocytes) are counted
- The primary value of the RBC count is that it allows determination of MCV and MCH

mean corpuscular
volume

Mean corpuscular values(MCV)

- Mean Corpuscular Volume (MCV)
 $(\text{Hct} \times 10) / \text{rbc}(\text{millions}) = \text{femtoliters}(\text{fl})$
Ex: pcv-45%, rbc count- 5 million/ul
 $450 / 5.0 = 90 \text{ fl}$
 - Increased in macrocytic anemias (B12, folic acid deficiencies)
 - Reticulocytosis causes transitory increase
 - Iron deficiency decreased MCV

Mean corpuscular values(MCV)

- Mean Corpuscular Hemoglobin(MCH)
 $(\text{Hb} \times 10) / \text{rbc}(\text{millions}) = \text{picograms}(\text{pg})$
Ex. Hb-15 gm, rbc count – 5 million/ul
 $150 / 5.0 = 30 \text{ pg}$
 - Decrease in iron deficiencies
 - Slightly increase/normal in reticulocytosis
 - Increase in vitro and vivo during hemolysis

Mean corpuscular values(MCV)

- Mean Corpuscular Hemoglobin(MCHC)
(Hb x 100)/Hct=grams per deciliter (gm/dL)

Ex. Hb-15 gm/dL ,pcv- 50%

$15 \times 100 / 50 = 30 \text{ gm/dL}$

- most accurate rbc indices
- reticulocytosis and iron deficiencies decreased

MCHC

Increase in vivo and vitro during hemolysis and spherocytosis

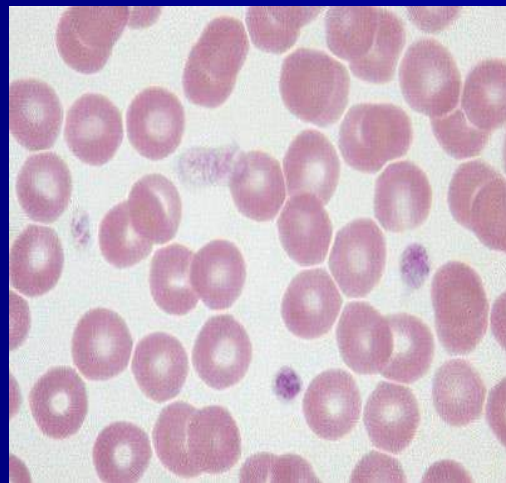
Sedimentation rate

- Principle: blood with anticoagulant is allowed to stand in perpendicular tube, erythrocyte sink because they are heavier than plasma which they are suspended
- Normal animals- speed of erythrocytes fall is slow and increased with anemia, inflammation and necrosis
- Most applicable to dog's blood

Species Differences for Erythrocytes

Canine erythrocytes

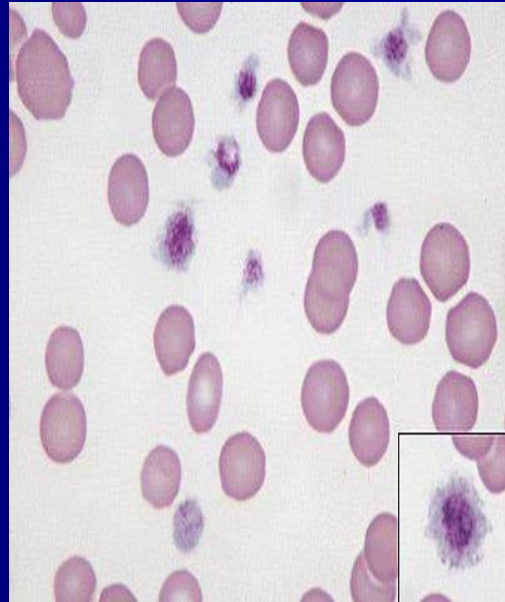
- large, uniform, biconcave disc with an area of central pallor
- Small numbers of polychromatophilic erythrocytes from healthy dogs (<1.5% reticulocytes)
- Occasional **nucleated** red blood cells and **Howell-jolly** bodies from healthy, non-anemic dogs
- lifespan varies from **110-120 days**



regenerative

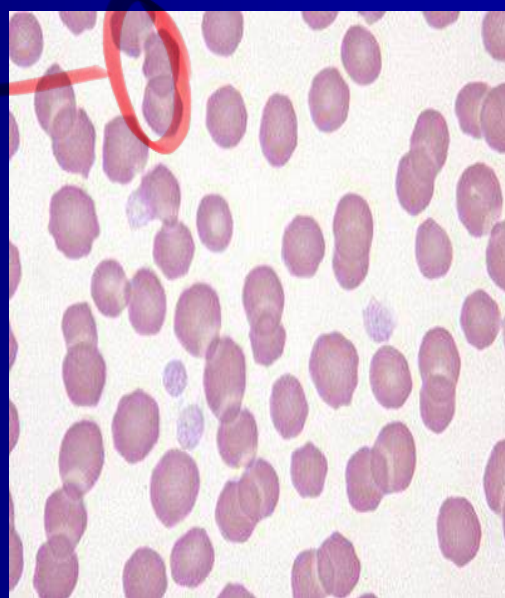
Feline erythrocytes

- smaller and more **variable** in size and shape than those of dogs
- **little to no** central pallor
- Polychromatophilic cells are very few in healthy (<1%)
- hemoglobin of cats is more susceptible to **oxidant injury** than in other species.
- low numbers (<10%)
single, **small heinz** bodies may be seen on erythrocytes in healthy cats
- **Healthy** cats also can have a few **Howell-jolly** bodies
- The lifespan of feline erythrocytes is **65-76** days



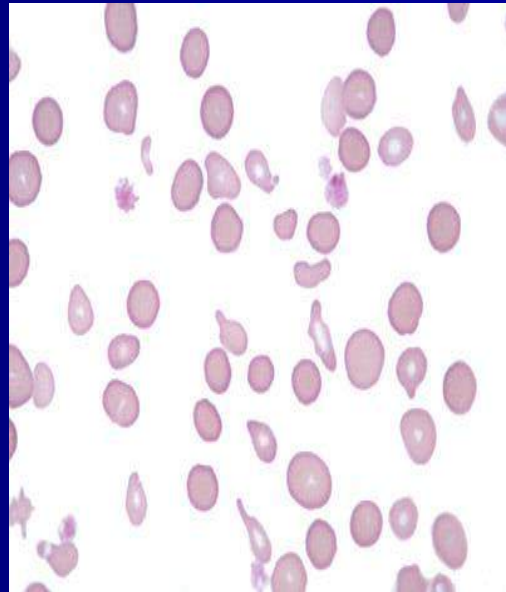
equine erythrocytes

- about the **same size** as feline erythrocytes and, similarly, **lack central pallor**
- prominent **rouleaux** formation
- Occasional **Howell-jolly** bodies
- Polychromatophils are **not** observed in blood from **healthy** horses and are **rarely** observed in blood from **anemic** horses
- **microcytosis** is a normal finding in **foals** up to **1 year** of age, and is attributable to a physiologic **iron deficiency** (**puppies** and **kittens** can display a similar **microcytosis** between **4-8 weeks** of age).



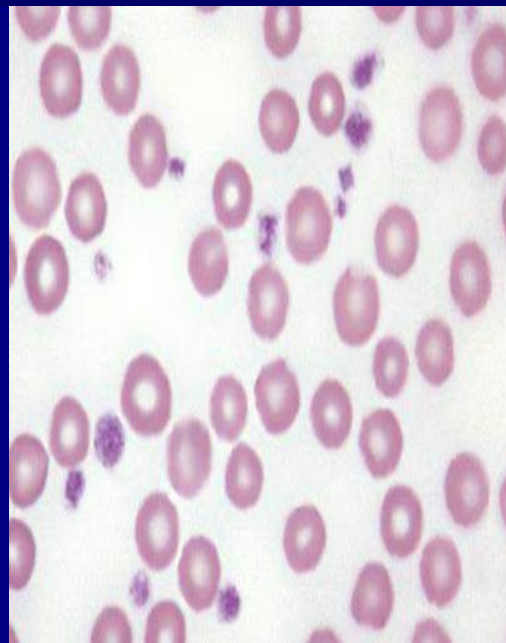
Caprine erythrocytes

- Marked **poikilocytosis** (especially **Angora**) and **prominent** in kids (< 3 months of age)
- **smallest**
- Do **not** display a prominent **reticulocytosis** in response to an **anemia** and lack **polychromasia** in peripheral blood in health



bovine erythrocytes

- similar in **size** to horse erythrocytes and lack **central pallor**
- A small degree of **anisocytosis**
- rarely display rouleaux formation in health or disease
- lifespan **is 160 days**
- **Polychromatophils** are not usually observed in blood smears from **healthy** non-anemic cattle, although they do release **reticulocytes** in response to an **anemia** (their capacity for **regeneration** is not as marked as the dog)



ERYTHROCYTE ABNORMALITIES AND INTERPRETATION

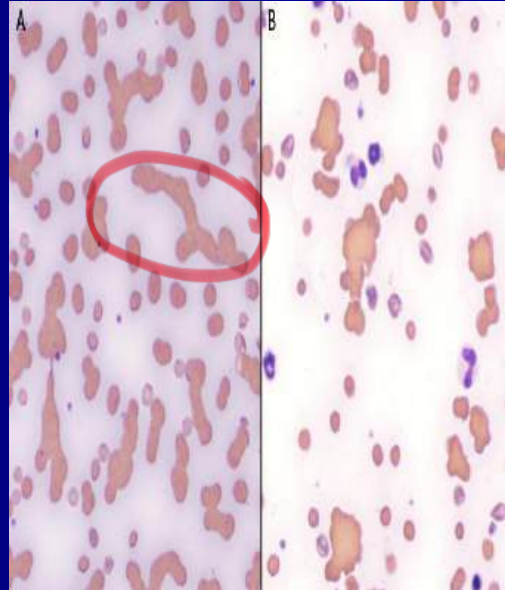
Rouleaux formation (pattern)

- – resembling stacks of coin
- parallel to ESR; common in healthy horses but not in cachectic or anemic
- mild in dogs and cats during inflammation and neoplasm
- rare in ruminants in health and disease



Agglutination (pattern)

- the phenomenon where red blood cells stick together in three-dimensional clumps
- due to the binding of antibodies to red blood cells
- When single antibodies bind to more than one blood cell, agglutinates form.
- Agglutination is usually a pathologic finding and supportive of a diagnosis of immune-mediated hemolytic anemia

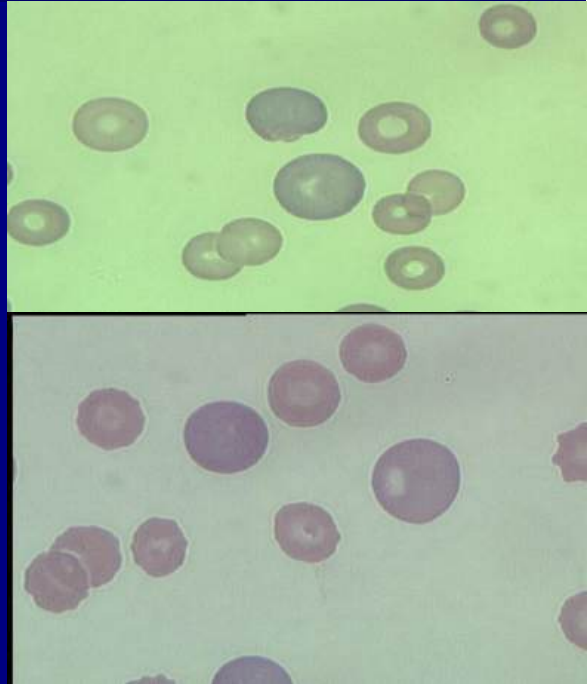


Agglutination vs Rouleaux

- distinguished from rouleaux by their characteristic appearance on blood smears however this can be difficult with severe rouleaux
- best way to distinguish between these two phenomena is to perform a saline dilution test, which is also called a saline dispersion test
- (https://www.youtube.com/watch?v=p4wF_5DkbOw)
- Saline dilution will disperse rouleaux but not agglutination

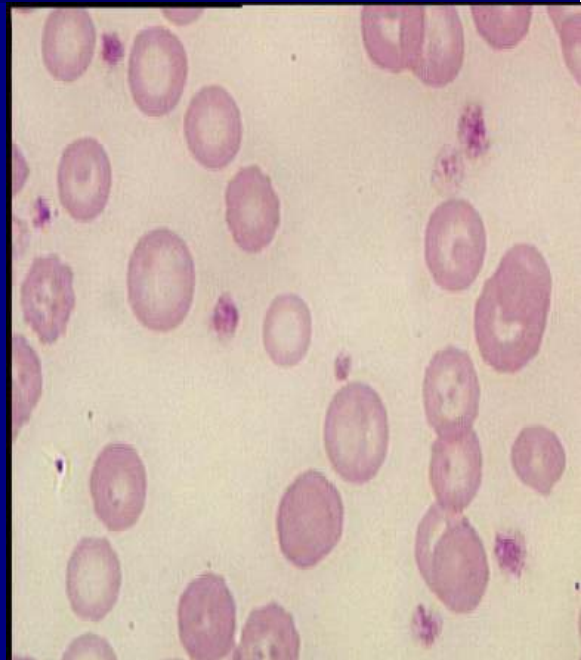
Anisocytosis (size)

- due to increased numbers of large RBC or small RBC, or a combination of both
- Some degree of anisocytosis is normal in animals; cattle then sheep, goat, cats
- this is the smear equivalent of the red blood cell distribution width (RDW) in automated machine
- **Result reporting:** Subjectively graded as mild, moderate or marked.



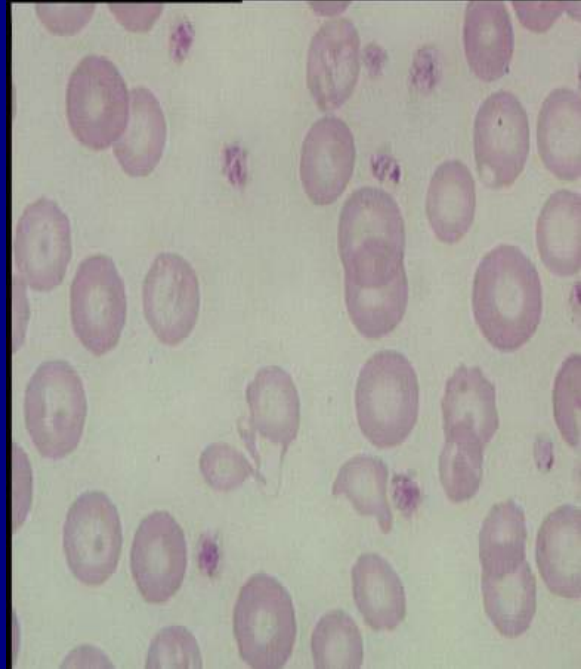
Microcytes (size)

- small cells with decreased MCV observed in iron and pyridoxine anemias
- represent cell remnants in heinz bodies and fragmentation anemias



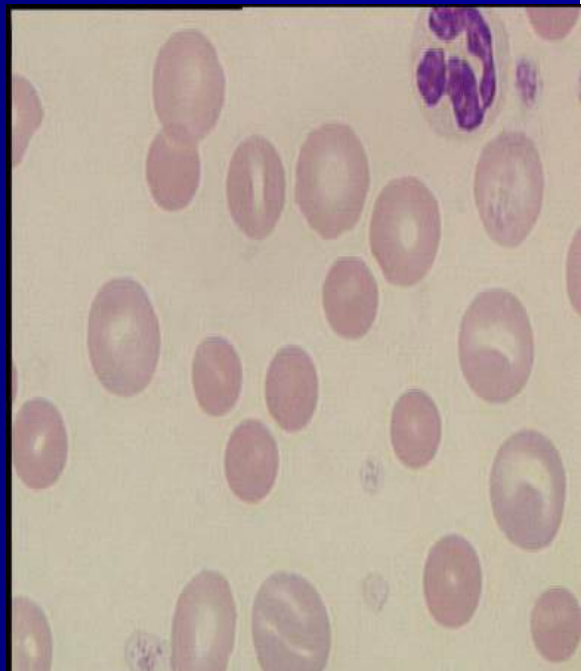
Poikilocytes (shape)

- Poikilocytes – abnormal shaped cells observed in chronic blood loss



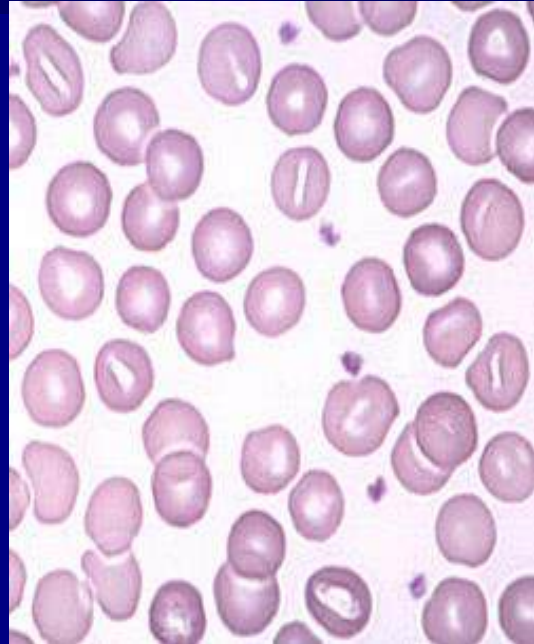
Spherocytes (shape)

- Spherocytes – small dark microcytes lacking central pallor
- readily detected only in dog and observed in autoimmune and isoimmune hemolytic anemias following transfusions



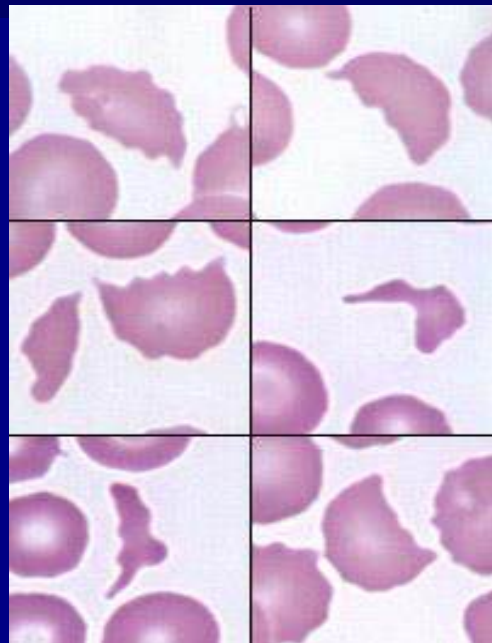
stomatocytes(shape)

- elongated (mouth-like) area of central pallor.
- occasional cell of this type might be seen as a non-specific finding in a variety of situations, such as regenerative anemias, liver disease, and lead poisoning
- an artifact in a blood smear that is too thick
- a marker of inherited red blood cell defects in dogs
- hereditary stomatocytosis some Alaskan Malamute dogs in association with chondrodysplasia, manifesting as dwarfism



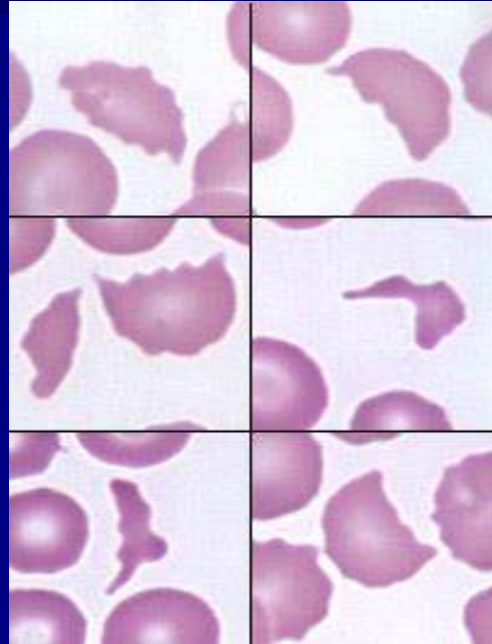
Schistocytes (shape)

- red blood cell fragments, are generally taken to reflect mechanical injury to erythrocytes.
- wide variety of forms may be observed
- Usually seen in relatively low numbers, they are associated with conditions in which the normally smooth endothelial lining is roughened or irregular, the vascular lumen is crossed by strands of fibrin, or blood flow is turbulent.



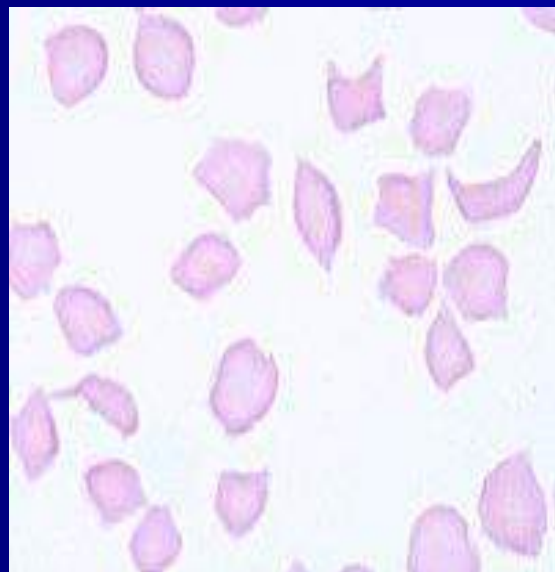
Schistocytes (shape)

- Disseminated intravascular coagulation, glomerular disease, vasculitis, portosystemic shunts and vascular neoplasms (e.g. hemangiosarcoma) are among the many conditions in which red blood cell fragmentation may be seen in varying degrees
- seen in up to 60-75% of dogs, but are rarely seen in cats and other species, with disseminated intravascular coagulation)
- Schistocytes also are commonly seen as a feature of iron deficiency anemia,



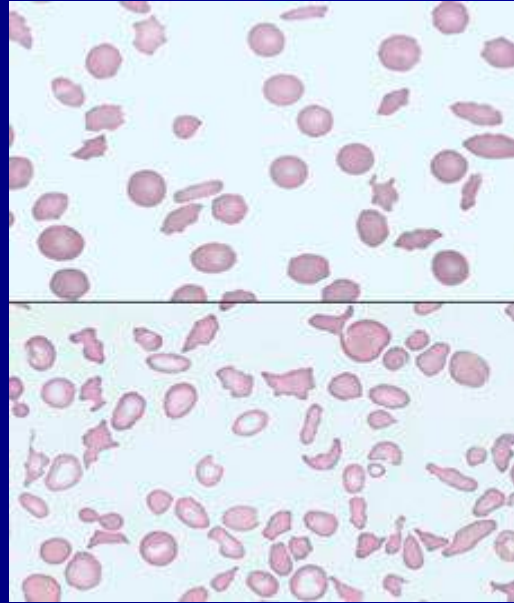
Drepanocytes(shape)

- A sickle-shaped erythrocytes that occurs in certain deer breeds (excluding reindeer and Montjac deer), antelope, some breeds of sheep (e.g. Barbary), goats, antelope, mongoose and genet
- sickling in deer (and probably other species) is an in vitro phenomenon due to variants of hemoglobin



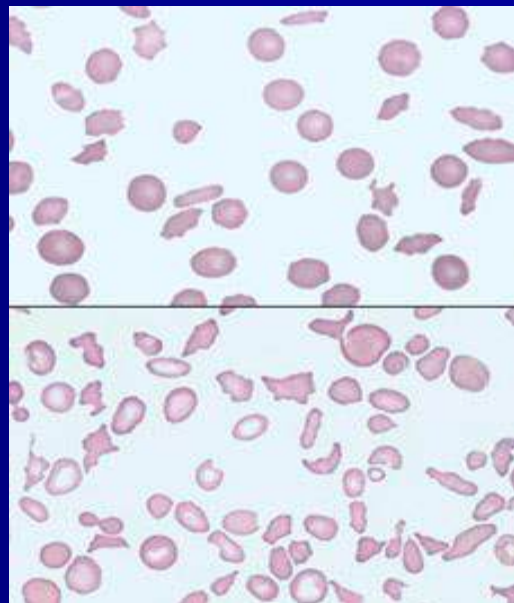
Elliptocytes (shape)

- elongated red blood cells
- three types: Type I is a slightly oval-shaped cell (used to be called ovalocyte), type II is a more rounded to oval shaped cell and type III is an elongate elliptical cell
- Distinction between these three types is not of clinical relevance, however some forms occur more frequently in some diseases



Elliptocytes (shape)

- Many cats with liver disease (particularly lipidosis) have acanthocytes and type III elliptocytes
- dogs should raise suspicion of underlying myelofibrosis
- Chemotherapy: Elliptocytes have been observed in cats given doxorubicin



Keratocytes(shape)

- a blister-like vesicle, which may rupture, leaving a “bite-shaped” defect in the cell outline or one or two horn-like projections on the same side of the cell



Keratocytes(shape)

- Fragmentation injury: Keratocytes will usually accompany schistocytes and acanthocytes in this setting. Associated conditions include causes of microangiopathic hemolysis (disseminated intravascular coagulation, vasculitis, hemangiosarcoma) and mechanical fragility, e.g. iron deficiency anemia.



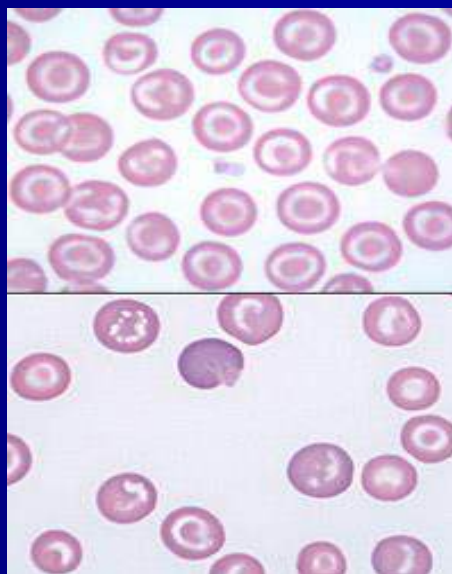
Keratocytes(shape)

- Oxidant injury: Here keratocytes may accompany eccentrocytes, pyknocytes, and possible Heinz bodies, depending on the oxidant.
- Liver disease: In cats, keratocytes can be seen in increased numbers in liver disease, e.g. hepatic lipidosis.



Leptocytes/target cells/codocytes (shape)

- Upper panel: Numerous target cells in the blood of a dog with liver disease.
- likely secondary to alterations in the lipid content or ratios of phospholipid: cholesterol of the RBC membrane, with expansion of the inner membrane leaflet.
- Dogs with liver disease can have target cells



Eccentrocytes (shape)

Hemighost
with
"eccentric"
central
pallor

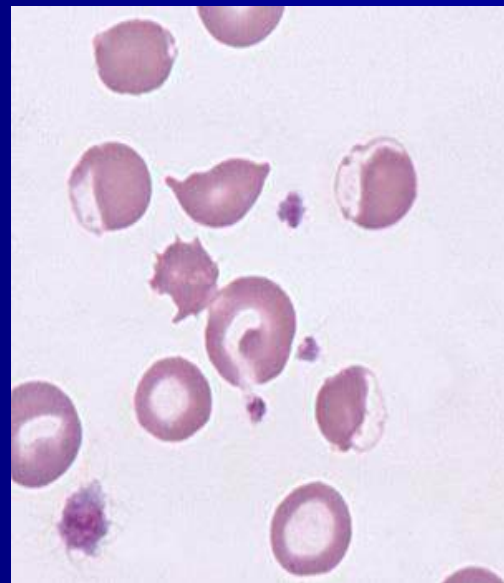
Mechanism: Oxidant-induced cross-linking of RBC membrane.

Drugs: Vitamin K1, phenothiazine drenches (horse), new methylene blue, propofol, acetaminophen (cats).

Disorders: Copper poisoning (sheep), red maple leaf toxicity (horses, camelids), *Pistacia* toxicity (horses), onions (dogs, cats, cattle, horses), *Brassica* species, e.g. kale (ruminants), zinc toxicity (dog), naphthalene in mothballs (dog), skunk musk (dogs, panda), inherited defects (glucose-6-phosphate dehydrogenase deficiency, flavin adenine dinucleotide deficiency; both in horses), various diseases in dogs associated with oxidant injury (lymphoma, diabetes mellitus, anticoagulant rodenticide toxicosis).

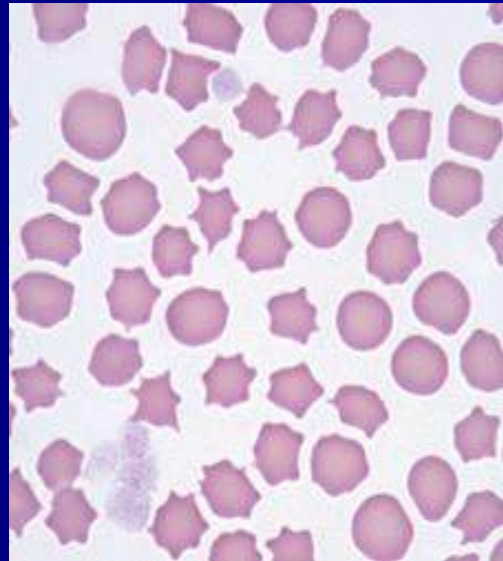
Eccentrocytes

- Three eccentrocytes from a dog with onion toxicity
- A shrunken echinocyte and a fragmented cell are present as well, probably also reflecting oxidant injury

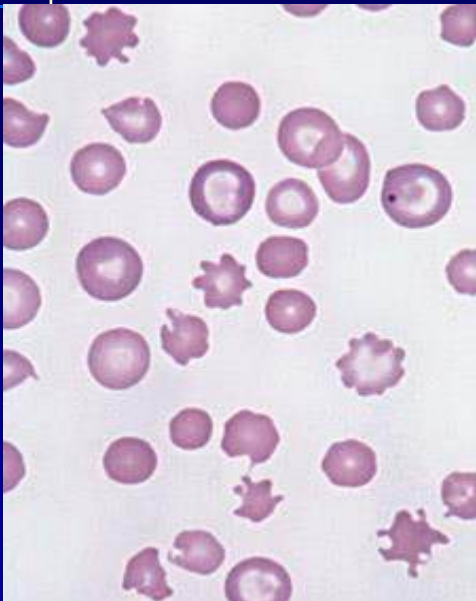


Echinocytes(shape)

- spiculated RBCs.
- term crenation (or “burr” cell) is also used to refer to cells of this type, particularly when the spicules are sharp and the echinocytes are an artifactual change
- projections of the cell membrane may be sharp or blunt, are usually numerous, and tend to be evenly spaced around the circumference.
- Spicules are usually of uniform size which differentiates echinocytes from acanthocytes (which have irregularly sized spicules, irregularly spaced around the cell membrane

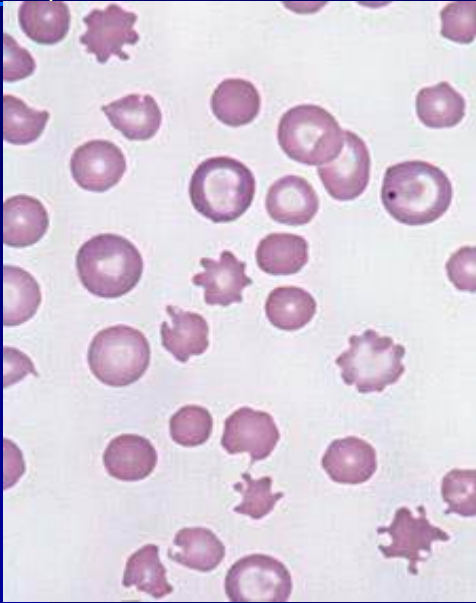


Acanthocytes (shape)



Irregularly spiculated RBC or “spur” cell

Acanthocytes



Mechanism: Mostly unknown, but speculated to be due to alterations in lipid composition of RBC membrane or fragmentation injury to RBCs

Physiologic: Young ruminants (calves).

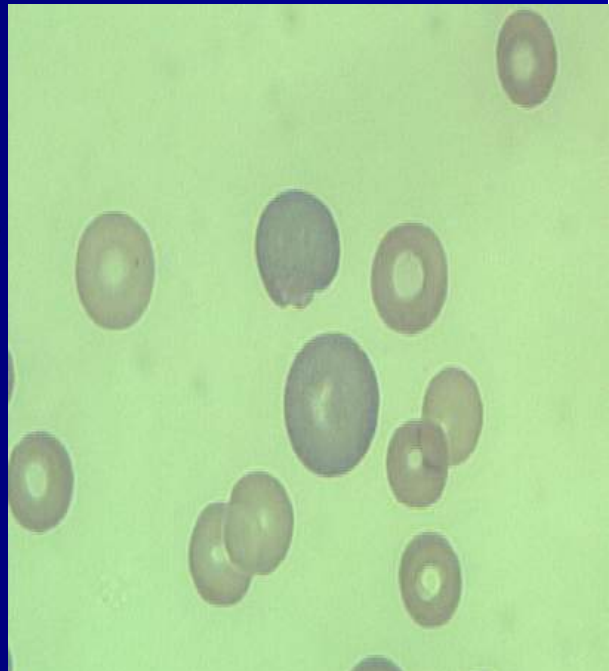
Disorders: Cancer, including hemangiosarcoma, (canine), liver disease (canine, feline), disseminated intravascular coagulation (DIC) (canine, bovine), vasculitis (canine), iron deficiency anemia (canine, mechanical fragility), possible congenital/inherited disorder (canine), various other diseases (e.g. renal, gastrointestinal, cardiac, canine).

Result reporting: Subjectively graded as few, moderate, many.

Differentiate from: Echinocytes (regularly spiculated). Can be difficult in individual animals.

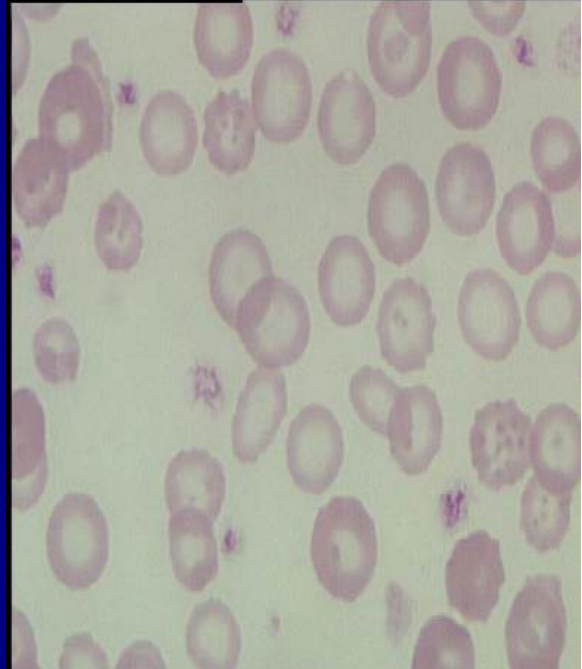
Polychromasia (color)

- Polychromasia
- variation in color among cells
- bluish erythrocytes are large and represent reticulocytes
- few polychromic cells are normal in dog and cat

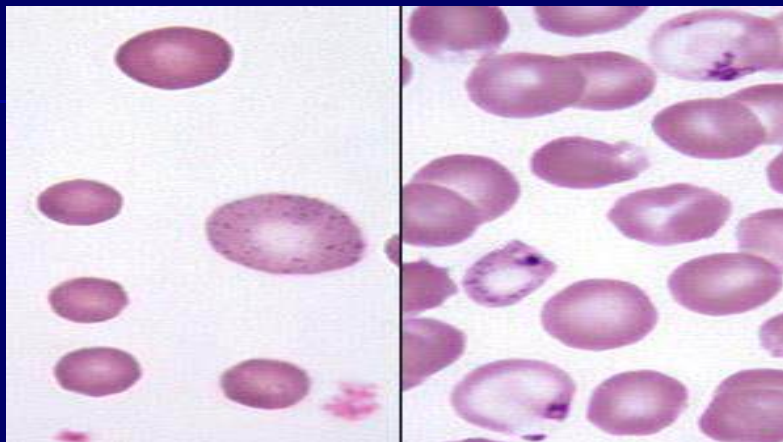


Hypochromia (color)

- Hypochromia – decreased staining intensity and increased central pallor due to insufficient hemoglobin; most common cause is iron deficiency



Basophilic stippling (inclusions)



RBC with chunky blue dots in a Romanow sky stain

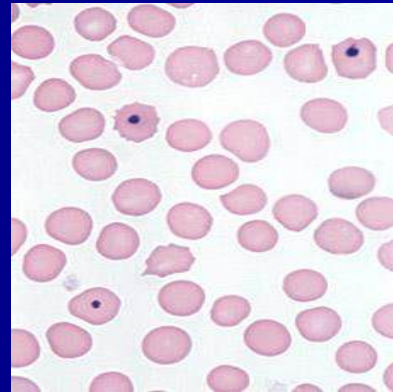
Mechanism: Aggregates of RNA due to RBC immaturity or inhibition of RNA degradation. Does not require intravital dye precipitation to observe in a regularly stained blood smear (Romanowsky stain).

Physiologic: Regenerative anemia (ruminants in particular, also dogs and cats but infrequent in the latter).

Pathologic: Lead poisoning (inhibits RNA degradation).

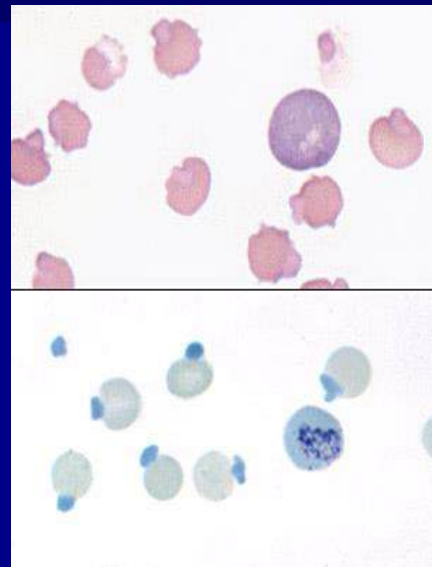
Howell – Jolly bodies (inclusions)

- (also called micronuclei) are small fragments of non-functional nucleus which were not extruded as the erythrocyte left the bone marrow.
- Usually, erythrocyte contains only a single Howell-Jolly body, which appears as a blue, perfectly round inclusion
- Occasionally, uneven staining of an Howell-Jolly body can result in a ring-like appearance which may mimic a parasite
- In dogs and ruminants, not routinely observed in peripheral blood because they are removed from red blood cells by the pitting function of the sinusoidal spleen.



Heinz bodies (inclusions)

- round structures on the internal erythrocyte membrane representing denatured hemoglobin
- caused by certain chemicals or oxidants



RED BLOOD CELL MORPHOLOGY					
Size variation	Hemoglobin distribution	Shape variation		Inclusions	Red cell distribution
Normal 	Hypochromia 1+ 	Target cell 	Acanthocyte 	Pappenheimer bodies (siderotic granules) 	Agglutination 
Microcyte 	2+ 	Spherocyte 	Helmet cell (fragmented cell) 	Cabot's ring 	
Macrocyte 	3+ 	Ovalocyte 	Schistocyte (fragmented cell) 	Basophilic stippling (coarse) 	Rouleaux 
Oval macrocyte 	4+ 	Stomatocyte 	Tear drop 	Howell-Jolly 	
Hypochromic macrocyte 	Polychromasia  (Reticulocyte)	Sickle cell 	Burr cell 	Crystal formation HbSC  HbC 	

QUICK GUIDE

- summary of terms used to describe red blood cell morphologic features, the meaning of the term and associated interpretation
- <https://eclinpath.com/hematology/morphologic-features/red-blood-cells/quick-guide/>

Anemia: Diagnosis and Classification

Anemia

- Absolute decrease in pcv, hemoglobin and rbc count
- Pcv is the easiest and most accurate in detection of anemia
- Classification: according to
 - size and hemoglobin concentration of erythrocytes
 - bone marrow response
 - Pathophysiologic mechanism

Anemia

hematocrit hemoglobin

- an absolute decrease in the Hct, Hb concentration, and/or RBC count
- ^{pcv} Hct is the easiest, most accurate method to detect anemia; value should be interpreted with knowledge of the patient's hydration status and with consideration of any possible influence of splenic contraction
- Hb concentration and RBC count may be used to further classify the anemia but usually are not needed to confirm its presence

detection

classification

$(Hb \times 100) / pcv$

mean corpuscular value

Classifications based on size(MCV) and Hb concentration (MCHC) of the erythrocyte

1. The MCV categorizes the anemia as normocytic, macrocytic, or microcytic. The average erythrocyte volume is within the reference interval, increased, or decreased, respectively
 norm macro micro
2. The MCHC categorizes the anemia as normochromic, hypochromic, or hyperchromic. The Hb concentration is within the reference interval if the erythrocytes are normochromic

hemoglobin

Hb concentration is decreased if the erythrocytes are hypochromic. Hyperchromasia is most often the result of erythrocyte hemolysis or Oxyglobin® administration, because erythrocytes do not overproduce hemoglobin

↓
↓↓

no over production, not true hyper

TABLE 2-9. Comparison of the Morphologic and Etiologic Classifications of Anemia

Morphologic Classification	Etiologic Classification
Normocytic normochromic	- Depression of <u>erythropoiesis</u> - Chronic inflammation - Nephritis with uremia <u>Endocrine deficiencies</u> (thyroid, adrenal medulla) <u>Neoplasia</u> <u>Marrow hypoplasia</u> as with Bracken fern poisoning, radiation, ehrlichiosis, chloramphenicol toxicity - Acute hemorrhage after fluid volume has been restored and before regeneration occurs - Feline leukemia virus infection
Macrocytic normochromic	- Dietary deficiencies - Vitamin B₁₂ deficiency - Folate deficiency

mCV
mCHC

↑ mCV

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Macrocytic normochromic	- Dietary deficiencies - Vitamin B₁₂ deficiency - Folate deficiency - Cobalt deficiency in ruminants - Erythremic myelosis in cats - Poodle macrocytosis
Macrocytic hypochromic	- During <u>recovery from large loss of RBC</u> <u>Hemorrhage</u> following injury or in animals with coagulation defects - Massive destruction of RBC as with immune-mediated anemia, hemoprotozoan infections, drug toxicity, congenital anemia of the basenji dog
Microcytic hypochromic	<u>Iron deficiencies</u> - Lack of <u>dietary iron</u> - Chronic blood loss to the exterior - Defect in <u>utilization of iron stores</u> - Copper deficiency - Molybdenum poisoning - Vitamin B₆ deficiency

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rocytes.

↑ mCV

Macrocytic hypochromic
↓ Hb conc.

↓ mCV

regenerating

Microcytic hypochromic
↓ Hb conc.

Classification according to bone marrow response

w/ response (bone marrow)

1. Regenerative anemia

a. The bone marrow actively responds to the anemia by increasing production of erythrocytes ↑

b. Laboratory Findings that denote regeneration of erythrocytes

(1) Polychromasia

RBC distribution

(2) Reticulocytosis with anisocytosis and increased RDW

(3) Macrocytosis (increased MCV) and hypochromasia (decreased MCH and MCHC) associated with reticulocytosis

(4) Basophilic stippling of erythrocytes in ruminants

(5) Hypercellular bone marrow with a decreased M:E ratio due to erythroid hyperplasia

C Classification according to bone marrow response

w/ response

2. Nonregenerative anemia

- suggests the lack of an erythroid response in the bone marrow
- Lack of response could be the result of inadequate time for erythropoiesis to occur as well as conditions such as chronic inflammation, renal disease, and endocrine disorders
- Horses do not release reticulocytes into blood; therefore, all anemias appear nonregenerative in this species

Classification according to bone marrow response

Non regenerative

- Polychromasia, reticulocytosis, and basophilic stippling (ruminants) are inadequate to absent
- During the first two to three days after the onset of peracute or acute hemorrhage or hemolysis, the anemia may appear nonregenerative
- Bone marrow examination occasionally may reveal the pathophysiologic mechanism or etiology of some cases of nonregenerative anemia

no response

Classification according to patho-physiologic mechanism

1. Blood loss (hemorrhagic anemia)
2. Accelerated erythrocyte destruction (decreased erythrocytic life span) by intra- or extravascular hemolysis/ Hemolytic anemia
3. Reduced or defective erythropoiesis/Bone marrow response

lysis

→ non regen?

Blood loss anemia(acute)

- Visual evidence of hemorrhage but may be occult (eg. Thrombocytopenia)

Laboratory findings:

- Normal pcv initially (blood are lost in similar proportion), may be hypovolemic shock → *pallor*
- Splenic contraction delivers high pcv(80%) to the circulation, temporarily elevating pcv ↑

Blood loss anemia(acute)

- Decreased plasma protein ↓ concentration
- ↑ ■ Increase platelet number during first few hours, thrombocytosis suggest continued bleeding ←
- Neutrophilic leukocytosis occurs 3 hours post hemorrhage

Blood loss anemia (chronic)

- Anemia develops slowly and hypovolemia does not occur

Laboratory findings:

- Pcv reach low values before clinical signs of anemia become obvious
- Regenerative response and hypoproteinemia
- Body stores of iron become depleted ↓
- Bone marrow response may end abruptly in prolonged cases

Does
not
show
early signs

Difference between external and internal hemorrhage

- External – prevents reutilization of ceratin components (iron, plasma protein) and reabsorbed followed internal hemorrhage (git is considered external hemorrhage)
- Internal – blood loss into body cavities; approximately 2/3 of the erythrocytes are reabsorbed into lymphatics (completed in 24-72 hours)

GIT

Causes of blood loss

acute	chronic
Trauma Surgery Large GIT ulcers Hemostasis defects - DIC, sweet clover poisoning	Parasitism Small GIT ulcers Hematuria Vascular neoplasia hemophilia

Hemolytic anemias

- Excessive destruction or shortened erythrocyte life span caused by variety of diseases such as parasites, bacteria, chemical agents, poisonous plants and metabolic diseases
- Blood parasites – anaplasma, piroplasma, eprythrozoon
- Bacteria – Leptospirosis and Bacillary hemoglobinuria

Hemolytic anemias

- Viral infection – EIA
- Chemical agents – lead, copper, phenothiazine, caponins, naphthalene, acetalinid, nitrofuratoln, some sulfonamides.
- Copper – sheep are the most susceptible to excessive intake
- Lead – normochromic erythrocytes, nucleated rbc, basophilic stippling and Howell-Jolly bodies
 - suspected in dogs reveals a significant number of immature erythrocytes

Hemolytic anemias

- Phenothiazine
 - anthelmintic accelerates the lytic action of naturally occurring lysolecithin present in horse blood
- Poisonous plants
 - castor beans, oak shoots, frosted turnips, broom ranunculus, ash, hazel (not appetizing to animals)
 - onions (wild and domestic) produce Heinz body hemolytic anemias, usually regenerative, macrocytosis and reticulocytosis.

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Bone Marrow Depression

- Physical agents – radiation
- Chemical agents – chloramphenicol, estrogens, phenylbutazone
- Chronic liver disease, hypothyroidism, hypopituitarism, parasitism, chronic nephritis

Bone Marrow Depression

Laboratory findings

- progressive fall in total erythrocyte count and hemoglobin content and a simultaneous decrease in total leukocyte count
- Close examination of hematocrit tube reveal clear, colorless and watery plasma
- Decrease thrombocyte
- Leptocyte may occur

normal blood volume

Polycythemia *high*

- Uncommon but may occur in polycythemia vera and hypoxia-stimulated erythropoiesis.
- Polycythemia vera – disease characterized by and absolute increase in total blood volume (increase in rbc count, pcv and hemoglobin content)
- reported in dog, cat and cow